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Optimal Charge Pattern for the High Performance Multi-Stage Constant Current Charge Method for the Li-Ion Batteries

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***Abstract*-- In this paper a method is introduced to find out the optimal charge pattern (OCP) of the multistage constant current (MSCC) charge method based on the equivalent circuit model of a Li-Ion battery. Unlike the conventional methods the proposed one does not require several kinds of time-consuming and labor-intensive experiments. The OCP can be simply calculated by the developed equations with the results obtained by a simple experiment. The validity of the proposed method has been verified through the experimental results and it shows that the charge time and the charge energy can be reduced by 12% and 1.8%, respectively, and the charge efficiency of a Li-Ion battery can be improved by 0.54 % as compared to the conventional CC-CV method. A lower temperature rise during the charge can be regarded as an additional advantage of the proposed method.**

***Index Terms*-- CC-CV method, equivalent circuit model of the battery, Li-Ion battery, multistage constant current charge, optimal charge pattern**.

I. INTRODUCTION

RECHARGEABLE batteries are becoming more popular in the energy storage solutions for the modern technologies such as portable electronics, Electric Vehicles (EV's), Plug-in Hybrid Electric Vehicles (PHEV's), Energy Storage Systems(ESSs) and renewable energy systems. Among them the Li-Ion batteries are being used dominantly now-a-days due to their several advantages including high energy density, long cycle life, high working cell voltage, lack of memory effect and low self-discharge rate. In addition, the Li-Ion batteries are robust to deep discharge, high charge rate (C-rate) and high ambient temperature which may affect their performance and life-span as well [1]-[9]. However, care must be taken to charge and discharge the Li-Ion batteries in order to obtain its optimal performance and longer life-span since it is strictly prohibited to overcharge and over discharge it. Therefore the fast and safe charge method without deteriorating the performance and life of the battery is highly desirable for the battery chargers.

The conventional Constant Current-Constant Voltage (CC-CV) charge is one of the most popular charge methods for the batteries due to its simplicity and effectiveness. It is composed of two kinds of different operations. During the CC mode charge a constant current is applied to the battery continuously until the terminal voltage of the battery reaches a certain maximum cut-off voltage, typically 4.2V in case of a Li-Ion battery. Although the terminal voltage has reached its cut-off value, the Electro-Motive Force (EMF) of the battery is still less than that of the terminal voltage because of the overpotentials caused by the internal impedance of the battery. In order to fully charge the battery, the EMF of the battery should be the same as the terminal voltage; hence the CV mode is applied to the battery. During the CV mode charge, the cut-off voltage is maintained across the battery and the current starts to decrease as the difference in voltages between terminal voltage and EMF decreases. When current reaches a certain value (typically 0.02C), the charge is terminated and the battery is considered to be fully charged. However, one disadvantage of the CC-CV method is that it takes relatively longer time in CV mode. Although more than 80% of the battery capacity is charged by the CC mode charge, about 50% of the total charge time is taken by the CV mode charge. The total charge time is a critical issue in the rechargeable battery applications such as EVs and PHEVs. It should be short enough in order for them to compete with conventional vehicles with internal combustion engines which can be refueled within 15 minutes typically. In order to shorten the charge time with CC-CV method, a higher charge current can be used in CC mode. However, only the charge time for the CC mode is shortened while the charge time for the CV mode is in fact prolonged due to the higher overpotentials caused by the internal impedance of the battery. Therefore, it would be a zero sum game. Furthermore, it is well known that a longer CV mode has adverse effects on the charge efficiency and the life-span of the battery due to its continuous structural stress [10].

Several kinds of different charge methods have been developed to shorten the charge time of the Li-Ion batteries [11]-[14]. In [11], a grey predicted charge method was proposed which can shorten the charge time by using a higher cut-off voltage during the CV mode. However, it was found that the higher cut off voltage has adverse effects on the life of the battery [10]. In [12], a sinusoidal-ripple-current (SRC) charge method was introduced. In this method a low frequency sinusoidal current superimposed on a DC current is used for the charge of the battery. Since the AC component of the charge current can be bypassed through the electric double layer

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capacitance of the equivalent impedance of the battery, total impedance of the battery can be minimized. Therefore a part of the losses associated with the charge transfer resistance can be reduced thereby improving the charge efficiency and lifetime of the battery. However, in [13] it was found that the real battery impedance is not minimized by using SRC charge strategy if the DC component of SRC is also considered. As a result SRC charge is not actually effective in reducing the charge time and temperature rise due to the increased RMS value of the charge current as compared to the CC-CV method. In [14], a variable frequency pulse charge method was proposed. Instead of using constant current, current pulses are used so that a rest period is provided for the ions in the battery to diffuse and to neutralize. The frequency of pulses is varied to reduce the battery impedance in order to improve the charge time of the battery. However, it requires a complex circuitry to generate the variable frequency pulses.

As an alternative to the conventional CC-CV method, Multistage Constant Current (MSCC) charge method has been proposed for the fast charge of the batteries [15]-[21]. Instead of using CV mode charge, it just uses several CC mode charges with different amplitudes of the currents. Therefore, the charge profile of the method is composed of several CC mode stages of which current amplitudes are different and the current at each stage is applied to the battery until the voltage of the battery reaches its cut-off value. It has been shown in [18] that the MSCC charge method has lower temperature rise as compared to CC-CV method due to its higher charge efficiency. In addition, it also helps increase the cycle life of the battery as it does not utilize the CV mode, which applies a constant stress to the battery. The performance of the MSCC charge method such as total capacity to be charged, charge time and charge efficiency depends on how to select the current amplitude at each stage. Hence, it is very important to find out the Optimal Charge Pattern (OCP) to optimize the charge performance with the MSCC charge method.

Many kinds of optimization techniques for MSCC charge method such as Ant Colony System Algorithm, Taguchi Approach, Fuzzy Logics and Particle Swarm Optimization have been proposed to find out the OCP of the MSCC charge method for the Li-Ion batteries [15]-[21]. However, in all of the above mentioned techniques, lots of pretests with various charge patterns are required to find out the OCP of the MSCC charge method as shown in Table I. Since these kinds of pretests are time-consuming and labor-intensive processes, it takes long time to find it and the overall cost of the system is increased. In addition the target capacity of the battery to be charged is not 100%, thus it is not practical to use these kinds of methods in the real applications as shown in Table I. In this paper it is proposed to find out the OCP of the MSCC method to charge the battery to its full capacity efficiently within a short period of time. In most of the previous researches, the authors only show the charge time and energy required to charge their batteries to their own target capacities such as 70%, 75%, 90% and 95% as shown in Table. I. Therefore it is impossible to make a fair comparison between the proposed method and the conventional methods as the charge time and the energy required to charge their battery do not vary linearly according to the capacity to be charged. Therefore we have compared the performance of the proposed method to those of the conventional CC-CV method. The proposed method has the following advantages:

1. The battery can be fully charged by the Optimal Charge Pattern (OCP) of the MSCC charge method obtained by the proposed method and the total charge time is shorter as compared to that of the conventional CC-CV method.
2. The OCP of any battery can be found with only one experiment by the proposed method. Hence, lots of tests to find out the OCP of MSCC charge method are not required, which can save a lot of time and efforts.
3. The energy required to charge the Li-Ion battery with the proposed method is less than that with the CC-CV method.
4. The temperature rise during the charge with the proposed method is less as compared to the CC-CV method and hence it helps increase the life-span of the battery.
5. The charge efficiency with the proposed method is higher than that with the CC-CV method.

TABLE I
COMPARISON BETWEEN THE PROPOSED METHOD AND THE CONVENTIONAL MSCC CHARGE METHODS

No.	Method Name	No. of tests required	Charge Capacity (%)	Charge Efficiency (%)*	Charge Time Reduction (%)**
1[17]	Fuzzy Logic	112	90	0.4	56.8
2[18]	Orthogonal Arrays	180	95	1.02	11.2
3[19]	Ant Colony	57	70	N/A	N/A
4[20]	Taguchi Method	40	75	0.96	N/A
5	Proposed Method	1	100	0.54	12

* Improvement in charge efficiency (%) w.r.t. CC-CV method
** Charge time reduction (%) w.r.t. CC-CV method

This paper is divided into four sections. Section I is the introduction. The proposed method is explained in detail in the section II. Section III shows the experimental setup and experimental results including the comparisons with the results obtained by the CC-CV method. Finally, the conclusion is given in the section IV.

II. Derivation of the Optimal Charge Pattern for the Multi-Stage Constant Current Charge Method for the Li-Ion Batteries

In this chapter, the detail of derivation of the OCP of MSCC charge method for the Li-Ion batteries has been presented. In the section A, a simple R-C equivalent circuit model of the Li-Ion battery is presented and the equations for calculating charge time (CT) and charge capacity by the MSCC charge method are derived. In the section B, the mathematical derivation of OCP by MSCC charge method is presented and finally the value of charge cut-off voltage in MSCC charge method is discussed in the section C.

A. Calculation of the Charge Time and the Charge Capacity

There are lots of kinds of equivalent circuit models of the Li-Ion battery. In order to focus on the dynamic behavior of the Li-Ion battery, an equivalent circuit model composed of multiple number of parallel R-C circuits is widely used in many literatures [22-25]. In many researches about the battery many kinds of different equivalent circuit models of the battery are

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used. Therefore a suitable equivalent circuit model of the battery should be selected and used depending on their applications. In order to compare the MSCC charge methods, only the overall charge time and the charge capacity are required to be calculated. Hence, a simple R-C equivalent circuit model can be used for modeling the battery for MSCC charge method. Although a simple R-C equivalent circuit model is not able to represent the fast dynamic behavior of the battery, it is enough to be used as a battery model to calculate the overall charge time and charge capacity. Since the proposed model is a kind of average model, it is not suitable to express the short term dynamic behavior of the battery. However, it is good enough to represent the long term behavior of it. Therefore, for the application of MSCC charge method a Li-Ion battery can be modelled by an equivalent circuit shown in Fig. 1 [26]. R_{eq} and C_{eq} denote the equivalent resistance and the equivalent capacitance of the Li-Ion battery, respectively. When the current I flows into the battery, its terminal voltage V_T can be given by (1).

$$V_T = V_{C_{eq}} + V_{R_{eq}} \tag{1}$$

khi có dòng vào
dòng vào điện trở là
vào luôn

dòng mất thời gian
nạp điện cho tụ
tuyến tính

$$V_{C_{eq}} = \frac{1}{C_{eq}} \int_{t_0}^{t} I(\tau) d\tau + V_s, \quad V_{R_{eq}} = I \times R_{eq} \tag{2}$$

Where, V_s represents the initial voltage across the equivalent capacitance C_{eq}. By combining (1) and (2), the terminal voltage of the battery V_T can be given as:

$$V_T = V_s + \frac{1}{C_{eq}} \int_{t_0}^{t} I(\tau) d\tau + V_{R_{eq}} \tag{3}$$

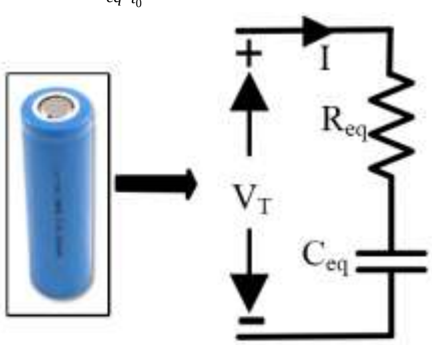

Fig. 1. An equivalent circuit model of a Li-Ion Battery

Since the maximum charge voltage $V_{T\_cut\text{-}off}$ of a Li-Ion battery can be found in its datasheet hence the time taken by the battery to reach $V_{T\_cut\text{-}off}$ from its initial condition can be calculated from (3) as (4).

$$\Delta t = \frac{1}{I}\left(V_{T\_\text{cut-off}} - V_s - V_{R_{eq}}\right) \times C_{eq} \tag{4}$$

In the MSCC charge method, (4) actually represents the charge time taken by a single current stage. Therefore, the total charge time T by the MSCC charge method can be represented as:

$$T = \sum_{x=1}^{n} \Delta t_x \tag{5}$$

Where,

$$\Delta t_x = \frac{1}{I_x}\left(V_{T\_\text{cut-off}} - V_{s,x} - I_x \times R_{eq}\right) \times C_{eq} \tag{6}$$

$$V_{s,x} = V_{T\_\text{cut-off}} - I_{x-1} \times R_{eq}, \quad x = 2, 3, 4\ldots \tag{7}$$

Where, x represents the stage number of the MSCC charge method. While, $V_{s,x}$ represents the voltage across the equivalent capacitor C_{eq} at the start of each stage. $V_{s,1}$ is the measured value of initial voltage across the battery before applying the first stage current while (7) can be used to find the initial voltage across C_{eq} before the start of next stages. Fig. 2 illustrates the MSCC charge method based on the equivalent circuit model of the Li-Ion battery.

The Ah capacity of the battery charged during each stage ΔAh_x can be calculated from (4) as (8).

$$\Delta Ah_x = \frac{1}{3600}\left(V_{T\_cut-off} - V_{s,x} - I_x \times R_{eq}\right) \times C_{eq}, \; x = 1, 2, 3\ldots \tag{8}$$

Hence, the total capacity of the battery Ah_T obtained by the MSCC charge method can be given as (9).

$$Ah_T = \sum_{x=1}^{n} \Delta Ah_x \tag{9}$$

Based on the equivalent circuit model of the battery, total charge time and capacity of the battery by MSCC charge method can be calculated by (5) and (9), respectively.

The value of equivalent capacitance C_{eq} can be calculated from the CC profile of a CC-CV method by using the charge-voltage relationship of the capacitor (Q=CV) shown in (4). Similarly, the equivalent resistance R_{eq} can be extracted from the time constant (τ_{eq}) of the battery, which can be calculated from the current profile during the CV mode of CC-CV charge and the previously calculated C_{eq} value given below [27]:

$$R_{eq} = \frac{\tau_{eq}}{C_{eq}} \tag{10}$$

In order to determine the optimal number of stages for the MSCC charge method the relationship between the number of stages and the charge performance has been investigated in [28]. It has been found that the charge performances in terms of charge time and charge efficiency are improved significantly until the number of stages reaches five. However, it is just slightly improved when the number of stages exceeds five. Therefore, it is not recommended to use higher number of stages than five because it would make control circuitry of the charger more complex without providing any prominent improvement in the charge performance.

It can be observed from (5) and (9) that in MSCC charge method, total charge capacity only depends upon the value of last stage current. Since the voltage drop across the R_{eq} resistor should be small enough in order to fully charge the battery, the last stage current needs to be selected carefully according to the desired charge capacity of the battery. Hence, for a five stage MSCC charge method, if the desired charge capacity of the battery is designated, then the value of last stage current I_5 can be calculated by using (9) and the values of Ah_T, R_{eq}, C_{eq}, $V_{T\_cut\text{-}off}$ and $V_{s,1}$ as in (11).

$$I_5 = \frac{1}{R_{eq}}\left(V_{T\_\text{cut-off}} - V_{s,1}\right) - \frac{3600}{C_{eq} R_{eq}} Ah_T \tag{11}$$

B. Derivation of the Optimal Charge Pattern for MSCC Charge Method

The first stage current I_1 in the MSCC charge method is in fact the same as that in the CC mode of CC-CV method. In case of the CC-CV method a charge current lower than the maximum allowable current value provided by the datasheet is

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applied to the battery during the CC mode charge until the battery voltage reaches a predetermined value of it. Thereafter the current is tapered by the CV mode charge until the current is smaller than a predetermined value of the current. The CC-CV method prevents the battery from overheating when the voltage of the battery approaches saturation and the battery can be fully charged by the tapered current within a reasonable amount of time.

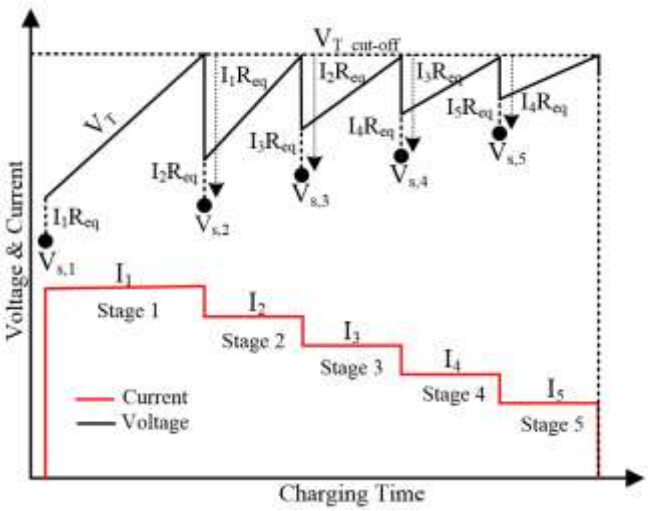

Fig. 2. Voltage and current waveforms of the MSCC charge method based on the equivalent circuit model of the Li-Ion battery.

However, in case of the MSCC charge method, since it is composed of several CC charge stages of which current values are different, a target charge capacity needs to be predetermined in order to determine the current value in the last CC charge stage. Since the first and last stage currents of the MSCC charge method can be determined if the maximum allowable current and the target charge capacity are known, the current values in the rest three charge modes need to be determined to optimize the charge performance.

Here, the derivation procedure for the OCP of the five stage MSCC charge method is presented. For a five stage MSCC charge method, the total time for the charge can be calculated by using (5)-(7) as in (12)-(13).

$$T = \Delta t_1 + \Delta t_2 + \Delta t_3 + \Delta t_4 + \Delta t_5 \tag{12}$$

$$T = \begin{cases} \dfrac{1}{I_1}\left(V_{T\_\text{cut-off}} - OCV - I_1 \times R_{eq}\right) \times C_{eq} \\ +\dfrac{1}{I_2}\left(I_1 - I_2\right) \times C_{eq}R_{eq} + \dfrac{1}{I_3}\left(I_2 - I_3\right) \times C_{eq}R_{eq} \\ +\dfrac{1}{I_4}\left(I_3 - I_4\right) \times C_{eq}R_{eq} + \dfrac{1}{I_5}\left(I_4 - I_5\right) \times C_{eq}R_{eq} \end{cases} \tag{13}$$

The optimized value of I_3, which is the middle stage current, can be calculated by taking the derivative of (13) with respect to I_3, putting it equal to zero and solving it for I_3 as shown in (14), (15) and (16).

$$\frac{dT}{dI_3} = 0 + 0 + \frac{-I_2 R_{eq} C_{eq}}{I_3^2} + \frac{R_{eq} C_{eq}}{I_4} \tag{14}$$

$$\frac{-I_2 \times C_{eq} R_{eq}}{I_3^2} + \frac{C_{eq} R_{eq}}{I_4} = 0 \tag{15}$$

$$I_3 = \sqrt{I_2 I_4} \tag{16}$$

Similarly the optimized values for I_2 and I_4 can be obtained by using the same procedure as (17) and (18):

$$I_2 = \sqrt{I_1 I_3} \tag{17}$$

$$I_4 = \sqrt{I_3 I_5} \tag{18}$$

By putting the values of I_2 and I_4 in (16) I_3 can be simplified as (19).

$$I_3 = \sqrt{I_1 I_5} \tag{19}$$

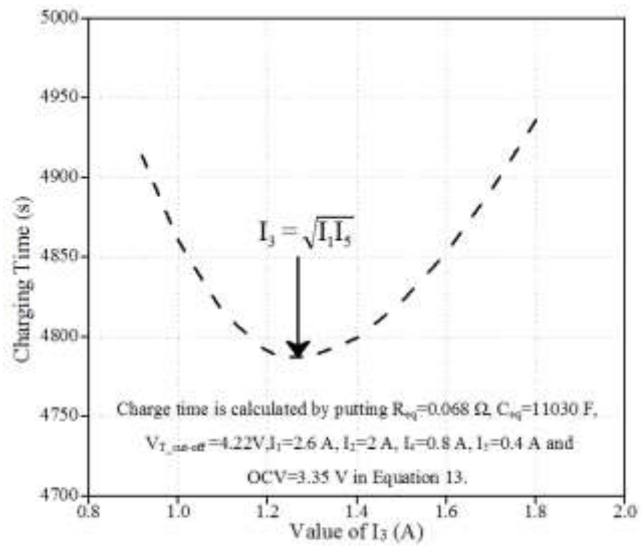

Fig. 3. Variation of the total charge time according to the value of I_3.

Fig. 3 shows the variation of total charge time T with respect to the variation of I_3 by using (13) and the parameters obtained in the section A. It can be observed from Fig. 3 that the total charge T is minimum when the I_3 has a value calculated by (19). It should be noted that the optimal value for I_3 does not depend upon the values of R_{eq} and C_{eq} instead it only depends on I_1 and I_5 values. The values of I_2 and I_4 obtained by using the similar procedure are also the optimized values which give the minimum charge time.

C. Effects of the Charge Cut-off Voltage

In most of the previous researches batteries were not charged to their full capacities by the MSCC charge methods. Hence it is not possible to make a fair comparison regarding the performances between the methods [17]-[20]. Furthermore it would not be an economic way to utilize the battery without charging it to its full capacity. Therefore the comparison by neglecting the charge capacity is less useful. For a fair comparison battery capacity charged by each method should be the same as the performance parameters are different depending on the charge capacities.

Unfortunately, however, it is almost impossible to charge the Lithium battery to its full capacity by the MSCC charge method within a reasonable amount of time since the last stage current becomes impractically small, which results in an even longer charge time than that of the conventional CC-CV method. Therefore, it is desirable to use a little bit elevated cut-off voltage to charge the battery to its full capacity within a reasonable amount of time.

TABLE II
RELATIONSHIP BETWEEN THE CHARGE CUT-OFF VOLTAGE AND THE DIFFERENCE IN CHARGE TIME W.R.T. CC-CV CHARGE METHOD

Current Pattern					Results from Calculations		
I_1	I_2	I_3	I_4	I_5	CT (s)	Voltage (V)*	Difference (%)**
2.6	1.01	0.40	0.15	0.06	7598	4.21	+28
2.6	1.45	0.81	0.45	0.25	5328	4.22	-12
2.6	1.57	0.95	0.58	0.35	4936	4.23	-17
2.6	1.72	1.14	0.93	0.5	4605	4.24	-22
2.6	1.84	1.3	0.92	0.65	4310	4.25	-27

*Charge Cut-off voltage
** Difference in charge time w.r.t. CC-CV method

For a five stage MSCC method, a relationship of charge cut-off voltage, the value of I_5 current and the reduction in charge time as compared to CC-CV method has been shown in Table II. For each cut-off voltage, the required value of I_5 current is different and can be found by using (11). It can be observed from Table II that if 4.21 V is used as a charge cut-off voltage in MSCC method then it will take 28 % more time than CC-CV method to fully charge the battery while if 4.22V is used then the charge time by MSCC charge method can be reduced by 12 % as compared to the CC-CV method. Similarly, by using 4.25V as a charge cut-off voltage in MSCC charge method, the total charge time can be reduced by 27 % as compared to CC-CV method. However, since 4.25V cut-off voltage may have some adverse effect on the lifetime of the battery, 4.217V (≈4.22V) has been used as a charge cut-off voltage in this research. The charge cut-off voltage is an important factor which may affect to the life of the batteries. However, it can be assumed that this small increase (just 0.02V more than that of conventional CV mode charge) will not have any significant adverse effect on the lifetime of the battery [11]. It can be realized from Fig. 6 in the section III.C that the average voltage applied to the battery during the CV mode of CC-CV method is higher than that during the last four stages charge by the MSCC charge method. In addition the time during which the battery voltage is higher than 4.2V is very small in the MSCC method with OCP obtained by the proposed method. Therefore it is difficult to judge which method is better in terms of safety and life of the battery and it can be considered as a good future research topic. However, the topic is beyond the scope of this research.

III. EXPERIMENTAL RESULTS

Fig. 4 shows the experimental setup for the proposed method which includes a bipolar power supply for the charge/discharge of the battery, a sensing circuit with Labview myDAQ for monitoring the results, a chamber to maintain the temperature at 25ºC and a desktop PC for applying the current profile and saving the results. A Li-Ion battery manufactured by LG (ICR18650 B4, 2600mAh) is used for the experiments.

In order to compare the charge/discharge capacity of the battery by using MSCC method and CC-CV method, two factors such as Normalized Discharge Capacity (NDC) and Normalized Charge Capacity (NCC) are introduced as (20) and (21), respectively.

$$NDC = \frac{C_{d-Multi}}{C_{d-CCCV}} \tag{20}$$

$$NCC = \frac{C_{c-Multi}}{C_{d-CCCV}} \tag{21}$$

Where, $C_{c\text{-}Multi}$ & $C_{d\text{-}Multi}$ denotes the charge and discharge capacities of the battery with MSCC charge method respectively while $C_{d\text{-}CCCV}$ is the discharge capacity of the battery by using CC-CV method.

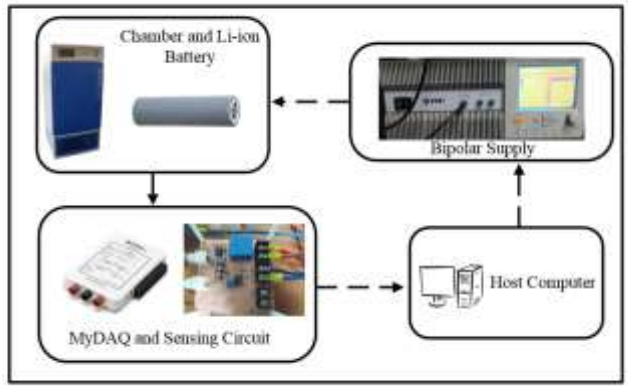

Fig. 4. Experimental setup for the battery test

A. Comparison of the calculated & experimental results of total charge time and total charge capacity

Based on the procedure given in section II, R_{eq} and C_{eq} of the battery equivalent circuit in Fig. 1 are calculated as 0.068Ω and 11030F, respectively. The value of the initial voltage across the battery before the test is 3.35 V. In this experiment the first stage current I_1 is selected to be 1C (2.6 A) as given in the datasheet of the battery. Here one thing should be noted that the values of R_{eq} and C_{eq} of the battery equivalent circuit may vary according to the ambient temperatures but the procedure to find the OCP will remain same. In this research the values of R_{eq} and C_{eq} are calculated at 25ºC temperature.

In Table III, a comparison of both calculated and experimental values of the total charge time (CT) and total charged capacity are given for the five stage MSCC method. The values of total charge time and total charge capacity are calculated by using (5) and (9), respectively. The comparison clearly shows that the maximum differences in total charge time and total charge capacity between the calculated values and the experimental results are 59 sec and 0.5%, respectively, which are negligible. The first current I_1 is the same for all of the experiments for five stage MSCC charge methods. It can be observed that the total charge capacity only depends upon the value of I_5 current while the discharge capacity depends upon the complete charge pattern as every charge pattern has different charge efficiency. The Fig. 5 shows the voltages obtained by the simulation and the experiments when a current

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TABLE III
EXPERIMENTAL & CALCULATED RESULTS OF VARIOUS 5 STAGE MSCC CHARGE METHODS

Charge Current Pattern					Charge Performance Obtained by the Experiments				Charge Performance obtained by the Calculations		Difference in Experiments and Calculations	
I_1	I_2	I_3	I_4	I_5	CT (s)	NCC (%)	NDC (%)	Efficiency (%)	CT (s)	NCC (%)	CT (s)	NCC (%)
2.6	1.8	1.2	0.8	0.4	4797	98.65	97.62	98.95	4774	98.52	23	0.13
2.6	2.1	1.8	1.1	0.6	4339	96.47	95.27	98.75	4347	96.93	8	0.46
2.6	2.1	1.4	0.9	0.5	4530	97.54	96.36	98.79	4511	97.72	19	0.18
2.6	2.2	1.5	1	0.45	4695	97.98	96.58	98.57	4719	98.12	24	0.14
2.6	1.5	1	0.8	0.3	5358	99.48	98.17	98.68	5303	99.31	53	0.17

TABLE IV
EXPERIMENTAL RESULTS FOR THE COMPARISON OF OPTIMAL CHARGE PATTERN

Charge Current Pattern					Performance of Each Charge Method			
I_1	I_2	I_3	I_4	I_5	CT (s)	NCC (%)	NDC (%)	Efficiency (%)
2.6	1.8	1	0.5	0.25	5364	100.50	99.62	99.12
2.6	1.5	0.9	0.6	0.25	5469	100.42	99.32	98.90
2.6	1.45	0.81	0.45	0.25	5224	100.55	99.95	99.40
2.6	2.2	1.8	1	0.25	5970	100.65	99.50	98.85
2.6	2	1.5	0.8	0.25	5663	100.62	99.78	99.16

TABLE V
PERFORMANCE COMPARISON BETWEEN PROPOSED 5 STAGE MSCC CHARGE METHOD AND CONVENTIONAL CC-CV METHOD

CC-CV Method	Charge Time (min)	98.83	Improvements
	Charge Capacity (Ah)	2.65	Charge Time (%) $= \frac{98.83-87.06}{98.83} \times 100\% = 11.90\%$
	Discharge Capacity (Ah)	2.62	
	Charge Efficiency (%)	98.86	
	Charge Energy (mWh)	10578	Charge Efficiency (%) $= 99.4\% - 98.86\% = 0.54\%$
MSCC Charge Method	Charge Time(min)	87.06	
	Charge Capacity (Ah)	2.6344	
	Discharge Capacity (Ah)	2.6186	Charge Energy (%) $= \frac{10578-10387}{10578} \times 100\% = 1.8\%$
	Charge Efficiency (%)	99.40	
	Charge Energy (mWh)	10387	

profile [2.6A, 1.45A, 0.81A, 0.45A and 0.25A] is applied on the battery. It can be clearly observed that the total charge time of both simulated and experimental results are quite close to each other as their difference in total charge time is about 50 seconds which is very small (0.1% difference). The results prove that the simple R-C equivalent circuit model used in this research is suitable for evaluating the performance of the MSCC charge method.

B. Verification of the Optimal Charge Pattern

When the first stage current I_1 is 2.6 A(1C rate), the last stage current I_5 can be calculated as 0.25A to charge the battery up to its 100% capacity by using 4.22V as a charge cut-off voltage based on (11). After deciding I_1 and I_5, the OCP can be calculated by using (17)–(19) as [2.6A, 1.45A, 0.81A, 0.45A and 0.25A]. In order to verify the performance of the OCP a comparison has been shown in Table IV. Several different charge patterns which have same values of I_1 and I_5 are applied to the battery to charge it to its full capacity. The cut-off voltage for all the MSCC patterns is (4.217≈4.22V). It can be observed that the optimal charge pattern is the best in terms of charge time (CT) and charge efficiency as compared to the other patterns as shown in Table IV.

C. Comparison of charge times with OCP of the 5 stage MSCC charge method and conventional CC-CV method

After finding out the optimal charge pattern, its performance is compared with that of the conventional CC-CV method. For the CC-CV charge of the battery, 2.6 A(1C rate) charge current is used for the CC mode and the CV mode is ended when the charge current reaches the value of 50 mA. The cut-off voltage for CC-CV method is 4.2V. Similarly 0.2C discharge current rate is used for calculating the discharge capacity of the battery with CC-CV and MSCC charge method, respectively.

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Fig. 6 shows the charge profiles of the battery with the conventional CC-CV method and the proposed 5 stage MSCC charge method. It takes 5930 seconds (98.83 minutes) with the conventional CC-CV method while it only takes 5224 seconds (87.06 minutes) with the proposed 5 stage MSCC charge method. Therefore, the battery can be charged up to 100 % capacity by the proposed method with 12 % less charge time as compared to the conventional CC-CV method.

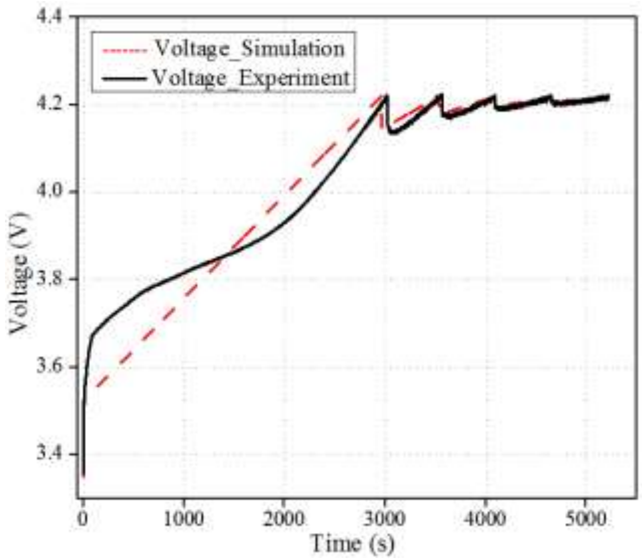

Fig. 5. Comparison of simulated and experimental voltage of MSCC charge method.

D. Comparison of the charge energy by OCP of the 5 stage MSCC charge method and conventional CC-CV method

The charge energies required to fully charge the battery by both methods are shown in Fig. 7. As the voltage of 5 stage MSCC charge method does not remain constant like the CV mode of CC-CV method therefore energy required to charge the battery by MSCC charge method is 1.8 % less as compared to that of CC-CV method.

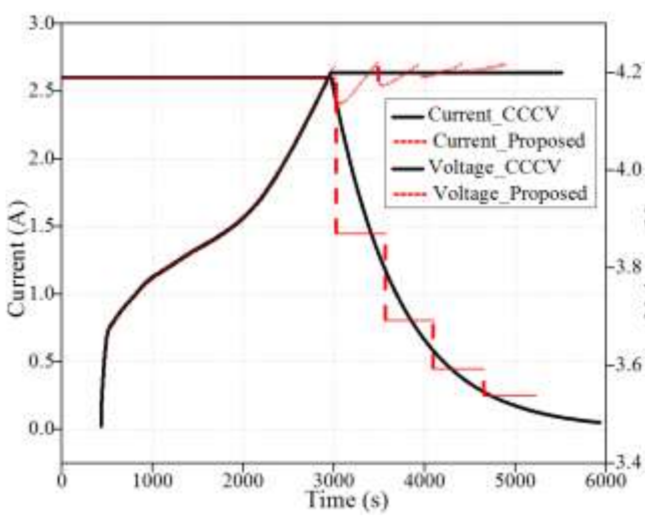

Fig. 6. Comparison of the CC-CV method and the proposed 5 stage MSCC charge method.

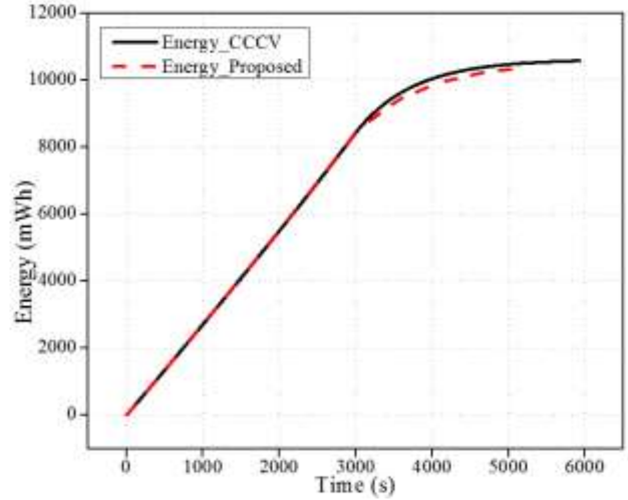

Fig. 7. Comparison of charge energy required by CC-CV method and proposed 5 stage MSCC charge method.

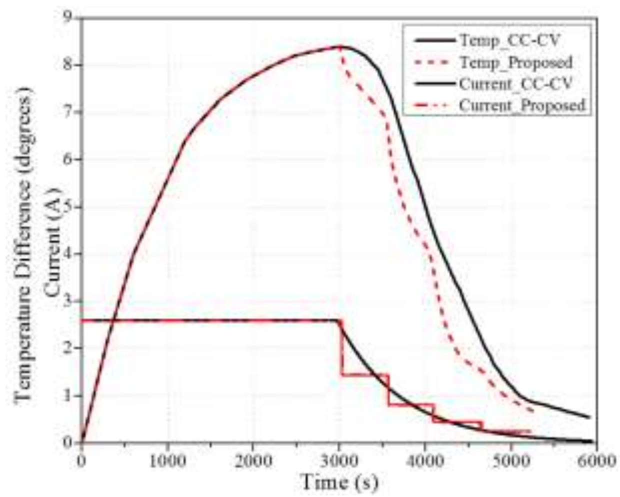

Fig. 8. Temperature variation of the battery during the charge by the CC-CV method and the proposed 5 stage MSCC charge method.

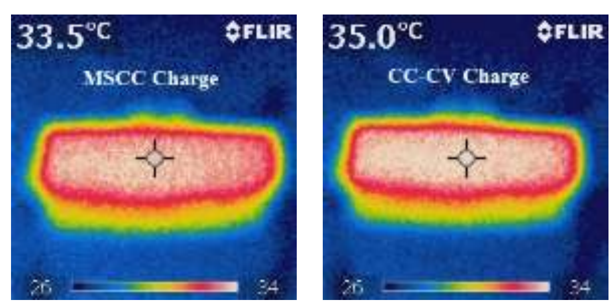

Fig. 9. Thermal image of the battery during the charge by the proposed 5 stage MSCC charge method and CC-CV method at t= 3650 sec in Fig. 8.

E. Temperature rise during the charge by OCP of the 5 stage MSCC charge method and conventional CC-CV method

The temperature rises during the charge by both CC-CV charge method and the proposed 5 stage MSCC charge method are shown in Fig. 8. Since the first stage current of the proposed 5 stage MSCC charge method is the same as the CC mode current of the CC-CV charge, the temperature rises in both

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methods are the same up to the end of CC mode of CC-CV method. However, the temperature variations in both methods are different after the CC mode charge. The temperature of the battery during the CV mode of CC-CV charge is higher than that of the 5 stage MSCC charge method as shown in Fig. 8. This can be considered as one of the significant advantages of the proposed method since the temperature rise during the charge affects the life time of the battery [29].

Fig. 9 shows the temperature of the battery during the charge by both methods at a specific time t=3650 sec in Fig. 8. The ambient temperature before the charge is 28ºC. The temperature rise with 5 stage MSCC charge method at t=3650 sec is 33.5 ºC while it is 35.0 ºC with CC-CV method. The lower temperature rise of the proposed 5 stage MSCC charge method can be considered as a proof for the higher charge efficiency of the proposed method as compared to that of the CC-CV method as shown in Table V.

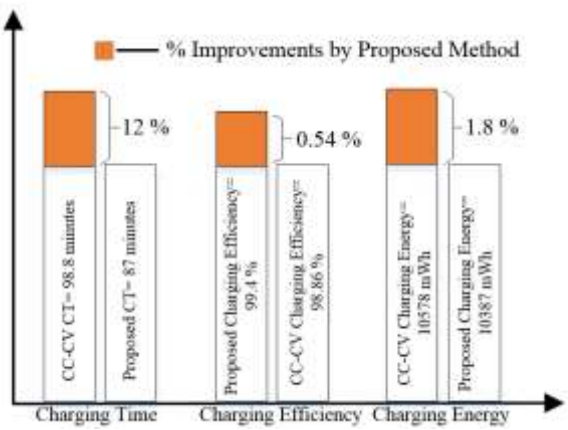

Fig. 10. Comparison of the charge performances by the proposed 5 stage MSCC charge method and CC-CV method.

IV. Conclusions

This paper has proposed a novel method to find the OCP for the 5 stage MSCC charge method for a Li-Ion battery. An equivalent circuit model of the lithium battery is used to derive the OCP for the proposed charge method and a method to extract the parameters of the battery are given in detail. Table V and Fig. 10 show the improvements in the charge performances by the proposed method as compared to those of CC-CV method. The experimental results show that the proposed method is able to charge the Li-Ion battery up to its 100 % capacity with 12 % less charge time as compared to the CC-CV method. The charge efficiency is improved by 0.54 % and the charge energy is reduced by 1.8 % with the proposed 5 stage MSCC charge method as compared to the CC-CV method. Unlike the other MSCC charge methods, the OCP of the MSCC charge method can be obtained without performing many kinds of pre-experiments. Therefore the labor and time for the pre-experiments can be saved significantly.

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